

ACADEMIC PROJECT ROADMAP

Cybersecurity Risk Management System

A focused plan for recording organizational assets, assessing cybersecurity risks, and tracking responsibility and status.

PROJECT TYPE	ACADEMIC CYBERSECURITY SYSTEM
CORE FOCUS	ASSET AND CYBERSECURITY RISK MANAGEMENT
DOCUMENT	PROJECT ROADMAP

Prepared as a clear implementation guide for the Cybersecurity Risk Management System project.

01. Project overview

This project allows an organization to record important assets, identify cybersecurity risks, calculate a basic risk level, assign responsibility, and track each risk through a clear status.

The problem

Security risks are often recorded in scattered documents or not tracked consistently. This makes priorities and responsibilities difficult to understand.

The response

A compact system will bring assets, risks, scores, owners, statuses, and summaries into one understandable workflow.

Project goal

Create a focused and complete cybersecurity risk-management workflow that is useful, academically worthy, and easy to understand. The application will support four essential activities:

- Record the assets that matter to the organization.
- Describe the cybersecurity risks that may affect those assets.
- Assess each risk consistently using likelihood and impact.
- Track responsibility, status, and the overall risk position.

Success principle

Every screen, field, and calculation must support the central asset-to-risk workflow. The project values clarity, correctness, and traceability over unnecessary complexity.

02. Users and navigation

The system uses two clearly separated user types. This keeps responsibilities understandable while providing enough access control for a meaningful academic system.

User	Primary responsibility	Main capabilities
Administrator	System and account administration	Manage accounts and use all project functions.
Risk User	Operational risk recording	Maintain assets, assess risks, and review summaries.

Visible navigation

The complete navigation contains five destinations. Each has one clear purpose and supports the main workflow.

Navigation item	Purpose
Dashboard	Quick view of assets, risks, open work, and high-priority risks.
Assets	Create, review, update, and archive organizational assets.
Risks	Create, assess, assign, filter, and track cybersecurity risks.
Reports	Review the risk summary and produce the printable risk register.
Profile	View the signed-in user and log out.

Accounts are created by the Administrator. The project does not require public registration.

03. Core feature set

The feature set is intentionally cohesive. Each area contributes directly to the same workflow, from asset registration to management reporting.

Area	Included options	Expected result
Authentication	Login; logout/profile	Users enter the system securely and operate under the correct role.
Dashboard	Four summary cards; one risk-level chart	The current risk position is visible at a glance, including an empty-state view.
Assets	Add/edit; view/archive	Important organizational assets are recorded with consistent ownership and importance.
Risks	Add/edit; view/filter/archive	Risks are connected to assets, assessed, assigned, and tracked.
Reports	Filtered on-screen summary; printable register	Risk information can be reviewed and presented in a consistent format.

Dashboard measures

Total assets Number of active assets.	Total risks Number of active risks.	Open risks Risks awaiting action.	High risks Risks requiring priority attention.
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The chart groups active risks into Low, Medium, and High levels. All measures must remain correct when no records exist.

04. Asset and risk information

The project records only the information needed to understand an asset and assess a related cybersecurity risk.

Asset record

Field	Purpose
Asset ID	Automatically assigned reference.
Name	Clear and recognizable asset name.
Description	What the asset is or what it supports.
Type	Data, System, Device, or Service.
Owner	Person responsible for the asset.
Importance	Low, Medium, or High business importance.
Active	Allows safe archiving without deleting history.

Risk record

Field	Purpose
Risk ID and title	Unique reference and short recognizable name.
Description and consequence	What may happen and the possible damage.
Asset	One affected organizational asset.
Likelihood and impact	Controlled values from 1 to 3.
Score and level	Read-only results calculated by the system.
Owner and status	Responsibility and current progress.
Notes and active state	Optional progress context and safe archiving.

05. Risk calculation and workflow

The assessment uses one transparent formula that can be checked manually and tested automatically.

$$\text{RISK SCORE} = \text{LIKELIHOOD} \times \text{IMPACT}$$

Score	Risk level	Interpretation
1–2	Low	Monitor through normal review.
3–4	Medium	Assign attention and track progress.
6–9	High	Prioritize action and ownership.

Users select likelihood and impact. The system calculates the score and level; those results cannot be entered manually.

Complete workflow

	Login
	View dashboard
	Add asset
	Create asset risk
	Select likelihood and impact
	Calculate score and level
	Assign owner and status
	Review risk register
	View or print summary

06. Implementation roadmap

	Stable foundation	• Confirm the approved project scope and requirements. • Create a reproducible development environment. • Record installation issues before implementation begins.
	User roles and navigation	• Create the project core module. • Add Administrator and Risk User roles. • Create the five navigation sections and verify intended access.
	Assets	• Implement the asset form and list. • Validate required fields and safe archiving. • Test access for both roles.
	Risks	• Implement the risk form and register. • Link risks to assets and add owner/status. • Add and test score and level calculation.
	Dashboard and report	• Add four summary cards and one chart. • Add filters and a printable risk register. • Verify every total against stored records.
	Documentation and quality	• Apply project branding. • Complete README, attribution, setup instructions, and change log. • Review clarity and define workflow tests.

07. Completion criteria

The project is complete when every item below has been implemented, tested, and documented.

☐ The application starts using documented steps.	☐ The approved project scope and requirements are documented.
☐ Login and logout work.	☐ Administrator and Risk User permissions work.
☐ Only the five approved navigation items appear.	☐ Assets can be added, viewed, edited, and archived.
☐ Risks can be added, viewed, edited, filtered, and archived.	☐ Every risk can be linked to an asset.
☐ Likelihood and impact accept only values from 1 to 3.	☐ Score and level calculations are always correct.
☐ Risk owner and status work.	☐ Dashboard totals and chart are accurate.
☐ The printable risk register works.	☐ Calculation, permissions, and the main workflow have automated tests.
☐ Every visible field and action can be explained clearly.	

Scope rule

If a proposed addition is not required by the project goal, user flow, or completion criteria, it does not belong in this roadmap.